PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2015

Candidate Name	

Civics Group						**INDEX NUMBER**					

H2 BIOLOGY **9648 / 03**

21 Sep 2015 **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.
This document consists of 2 sections:

Section A
Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B
Answer Question 5.
Write your answer on the lines provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	**/ 13**
2	**/ 14**
3	**/ 13**
Section B	
5	**/ 20**
TOTAL	**/ 60**
4 (SPA)	**/12**

This document consists of **17** printed pages (inclusive of this page).

Question 1 [13 marks]

Restriction enzymes cut DNA molecules only at specific target sites with particular base sequences. The target sites for the enzymes *EcoR*I, *BamH*I and *Hpa*I are shown in **Table 1.1**. The lines drawn in each sequence show where the enzyme cuts the DNA molecule.

Table 1.1

restriction enzyme	specific target base sequence of DNA
*EcoR*I	G | A A T T C C T T A A | G
*BamH*I	G | G A T C C C C T A G | G
*Hpa*I	G T T | A A C C A A | T T G

A bacterial plasmid includes the base sequence shown in **Fig. 1.1**. There are no target sequences for these three restriction enzymes in the rest of the plasmid.

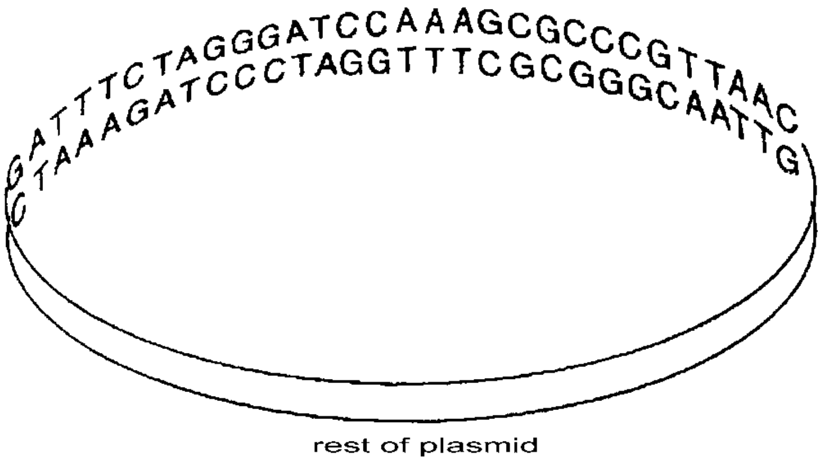


Fig. 1.1

a) State how many restriction fragments of DNA will be present after the plasmid has been treated with.

i. *Hpa*I and *EcoR*I: [1]

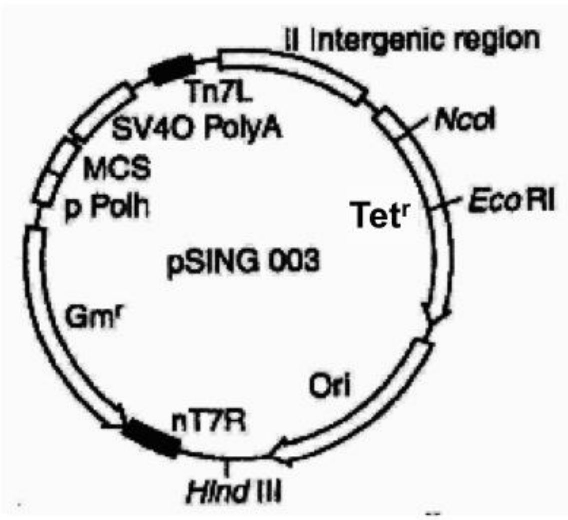
1 fragment

ii. *Hpa*I and *BamH*I: [1]

2 fragments

Antithrombin deficiency (AD) is a blood clotting disorder in which sufferers produce insufficient amounts of antithrombin. Antithrombin binds to thrombin and herparin resulting in deactivation of the blood coagulation pathway. AD results in formation of blood clots in blood vessels which obstruct blood flow in the circulatory system.

Fig. 1.2 shows a vector map of the plasmid vector pSING003. The plasmid is used to clone the antithrombin gene, which will be inserted at the *EcoR*I site. The plasmid is subsequently introduced into *E. coli* via transformation. After transformation, the *E. coli* cells are then grown on an agar plate.



Gm^r – gene coding for resistance to the antibiotic gentamycin

Tet^r – gene coding resistance to tetracycline antibiotic

Ori – origin of replication

Fig. 1.2

b) Describe the natural function of *Eco*RI restriction enzyme. [1]

a. protect bacteria from attack by <u>viruses/bacteriophages</u> ;
b. cleave any foreign DNA/hydrolyse viral DNA that enters bacterial cell / by cutting up intruding DNA from other organism

c) With reference to Fig. 1.2, state two of the most important considerations to make when choosing *EcoR*I as the restriction enzyme to use for the insertion of foreign gene into the plasmid. [2]

a. The EcoR1 restriction site (RS) is unique in the plasmid;;
b. The RS must be found within one of the two selectable markers within the plasmid;;

d) In addition to the nutrients required for growth, name **two** other substances that are added to the agar plate to allow selection of cells successfully transformed with the recombinant plasmid. [2]

a. Antibiotic gentamycin
b. Antibiotic tetracycline

Bacteria were then mixed with the recombinant plasmids. Those bacteria which had successfully taken up recombinant plasmids were identified using the following steps:

Step 1 – the bacteria were spread onto culture plates containing nutrient agar and gentamycin and incubated to allow colonies to form, as shown in Plate A in Fig. 1.3.

Step 2 – some bacteria from each of the colonies growing on Plate A were transferred to plates containing nutrient agar and tetracycline, as shown in Plate T in Fig. 1.3.

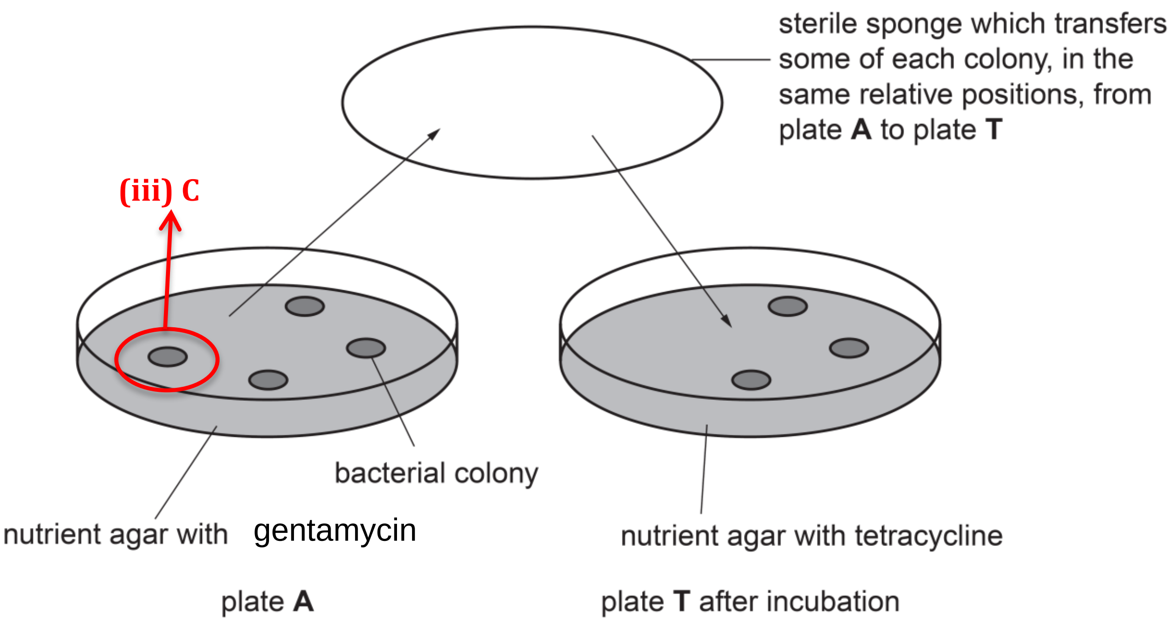


Fig. 1.3

e) Explain why bacteria must first be plated on gentamycin. [2]

a. identifying bacteria that are transformed
b. Untransformed bacteria these bacteria will be killed in this medium

f) With reference to Fig. 1.3, explain why it is important for *EcoR*I site to be within the tetracycline resistance gene. [2]

a. EcoR1 cuts at the RS within Tetr gene & inserts the gene of interest into the gene disrupts the tetracycline resistance gene -> Insertional inactivation of the Tetr gene
b. Colonies that are gentamycin resistant but not tetracycline resistant have taken up the recombinant plasmids.

g) Label ‘C’ on Fig. 1.3 to indicate the colony of bacteria that has taken up the recombinant plasmid. [1]

h) What is the role of the sterile sponge in Fig. 1.3? [1]

Replica plating

<u>Question 2 [14 marks]</u>

Restriction Fragment Length polymorphism (RFLP) analysis can be used as a method of detecting anti-thrombin deficiency (AD) in family members by using a radioactive probe to determine the presence of the mutant allele that results in AD.

a) Explain the principles of RFLP analysis. [1]

RFLP refers to the production of different number of DNA fragments of different lengths when genome from 2 different individuals are cleaved by the same restriction enzyme

Fig. 2.1 revealed two RFLP morphs that are tightly linked to the AD gene which can be used in the screening of AD. Both RFLP morphs and probe is shown in Fig. 2.1 with arrows representing the *EcoR*I restriction sites.

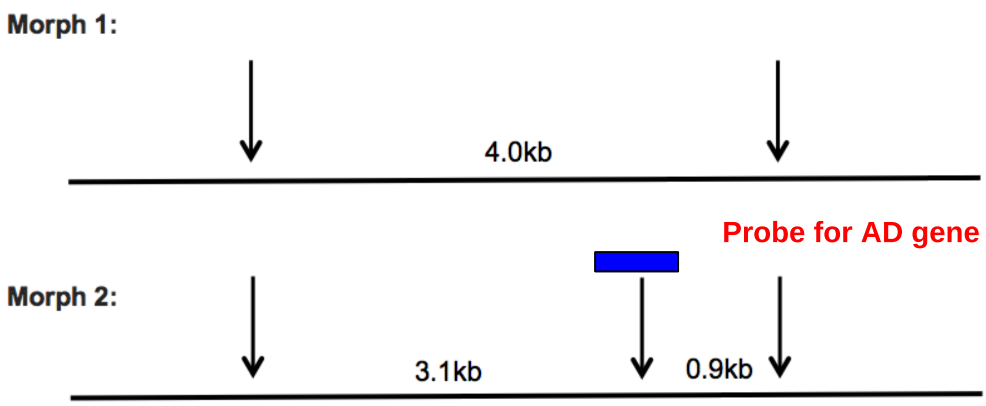


Fig. 2.1

Fig. 2.2 illustrates the Southern Blot analysis of both RFLP morphs in individuals with a family history of AD.

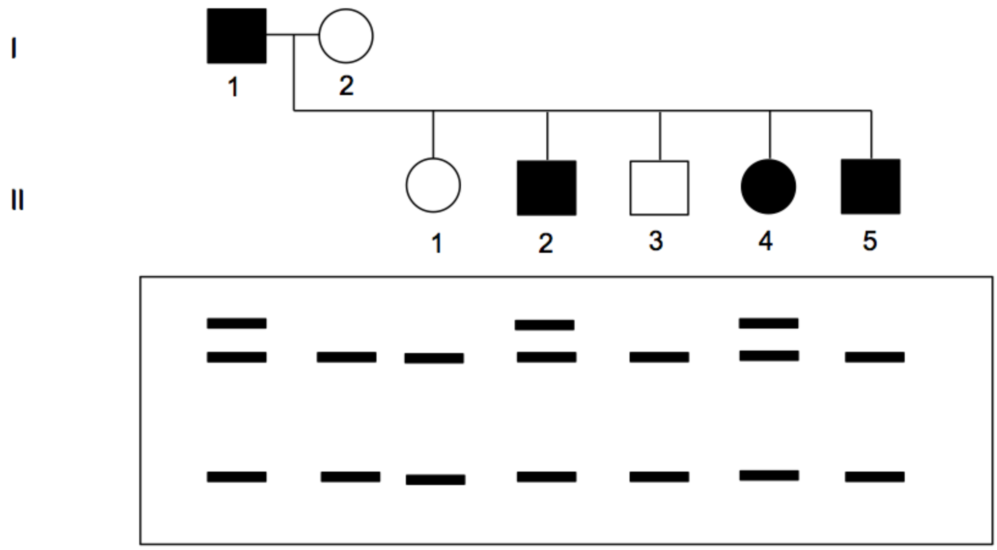


Fig. 2.2

b) Describe the process of obtaining a Southern Blot for disease detection. [3]

a. Cut genomic DNA using an appropriate RE(s) & carry out gel electrophoresis
b. Restriction fragments from the gel are then transferred onto a nitrocellulose membrane via capillary action.
c. Radioactive DNA probes specific for the polymorphic region / disease causing gene are allowed to bind to the DNA on the nitrocellulose membrane

c) On Fig. 2.1, draw the location of the probe that would produce an autoradiogram as indicated in Fig. 2.2. [1]

correct position of probe;
labelling;

d) With reference to Fig. 2.1 and Fig. 2.2, state and explain which RFLP morph is linked to the mutant AD gene. [3]

a. RFLP morph 1;
b. require 1 copy of the dominant allele for the individual to inherit the disease;
c. Only affected individuals, I1, II2 and II4 have a copy of RFLP morph 1 thus a band of 4.0kb is seen and;

e) Suggest the expected result of the Southern blot analysis of individual II5 and the possible reason for the difference in the DNA patterns for individual II5 from the other affected individuals. [2]

a. During gamete formation/meiosis, in one of the parents (I1);
b. Crossing over occurred at the loci of homologous chromosomes between RFLP genetic marker and AD mutant gene;

f) State two features of DNA probes that can be used to detect RFLP. [2]

a. DNA probes which are single stranded molecules;;
b. radioactively labeled;; OWTTE

g) Outline how one other disease can be detected by RFLP. [2]

a. **Sickle Cell anemia**
b. **Base-pair substitution**
c. **Loss of RS**
d. **Sickle cell allele – a larger fragment was obtained & Normal allele - presence of one restriction site, two shorter fragments**

Question 3 [13 marks]

Cystic fibrosis is a genetic disorder that affects mostly the lungs and also the pancreas, liver, kidneys and intestine. The average life expectancy of a cystic fibrosis patient is between 37 to 50 years old.

One of the treatments for cystic fibrosis (CF) is to inhale a mist of sterile hypertonic saline (an intensely salty solution).

a) Suggest how hypertonic saline can be an effective treatment for CF. [1]

Hypertonic saline inhalation increases ion concentrations in the airway surface liquid and osmotically draws water from epithelial cells into the airway lumen

Another treatment would be the use of gene therapy, which involves transferring normal CFTR alleles into lung epithelial cells with defective genes.

b) Describe how a non-viral gene delivery system can be used to insert the normal CFTR allele into a CF patient. [3]

a. **Gene coding for CFTR is isolated/obtained from the a normal cells of individual not suffering from CF/gene → plasmid**
b. **Liposomes (mainly cationic or positively-charged) are packaged with normal CFTR allele.**
c. **Lipoplex -> Cationic lipids, due to their positive charge, naturally complex with the negatively charged DNA**
d. **Sprayed into nasal tract**

In a clinical trial involving 16 patients suffering from cystic fibrosis (CF), liposome-mediated transfer of the normal CFTR gene was carried out using aerosol spray through the nose. Eight patients received the liposome containing normal CFTR gene, while the other 8 patients were given the same treatment without the normal CFTR gene (liposome alone). This latter group is designated as the placebo group.

One of the ways to study the effectiveness of the treatment was to measure the *in vivo* potential difference across the epithelial cell membrane after a low chloride concentration solution was passed through the airway of the lung. Results are shown in Fig. 3.1. Potential difference across a membrane is the measure of concentration of cations and anions inside and outside the cell.

The potential difference is measured 2 days pre-treatment and 2 days post-treatment for each patient and the mean is calculated. The mean potential difference seen in people without CF is also measured.

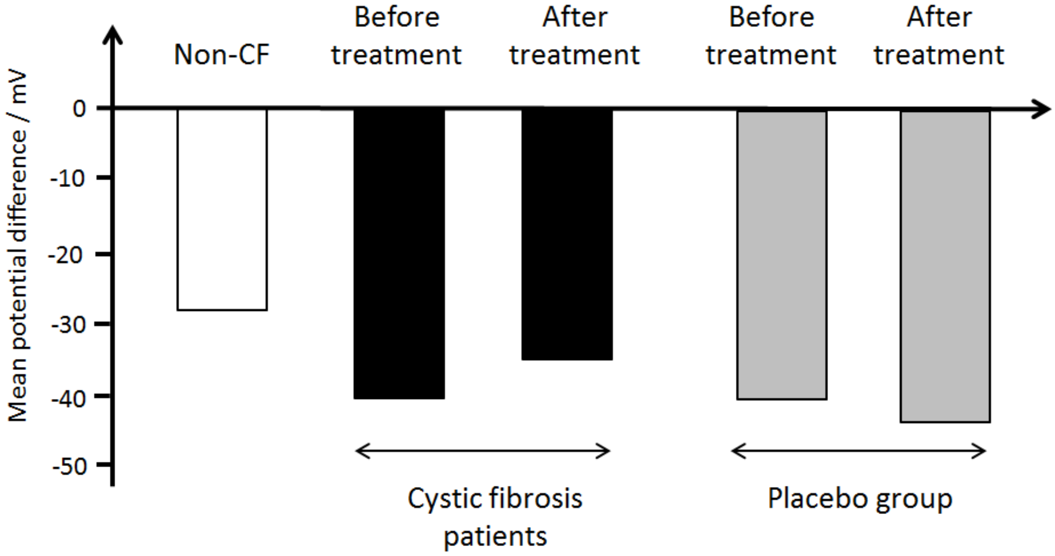


Fig. 3.1

c) With reference to Fig. 3.1, explain the evidence that shows gene therapy treatment for CF is successful. [3]

a. **Quote values and trend observed**
b. **This suggests that with the normal CFTR gene, the normal CFTR protein is produced and hence chloride channels are functional.**
c. **chloride ions are able to diffuse out of the epithelial cell / efflux of Cl^- io**

In liposome-mediated gene therapy of CF, the effectiveness of the treatment is very low, even though an immune response is not generated in patients. Many clinical trials, some of which are described in Table 3.1, have reported low effectiveness even though measurements were taken shortly after treatment.

Table 3.1

Clinical trial / year	Liposome type	No. of patients	Target	Efficacy
Caplen et al. / 1995	DC-Chol / DOPE	15	Nose	Potential difference 20% correction towards normal
Zabner et al. / 1997	GL-67™ / DOPE	12	Nose	Partial correction nasal potential difference
Porteous et al. / 1997	DOTAP	16	Nose	Potential difference 20% correction in 2 patients
Noone et al. / 2000	EDMPC-Chol GL-	11	Nose	None
Alton et al. / 1999	DOPE / DMPE-PEG_{5000}	16	Lungs	Potential difference 25% correction towards normal
Ruiz et al. / 2001	DOPE / DMPE-PEG_{5000}	8	Lungs	Vector-specific mRNA in 3 out of 8 patients, no functional protein detected

d) Suggest **two** reasons why effectiveness of treatment is low when cationic liposomes are used for CF gene therapy. [2]

a. The endosome formed after endocytosis of the cationic liposome is degraded by lysosomes in the cell. The normal functional gene is hence degraded.
b. Inability of the normal functional gene to exit the cationic liposome before entry into nucleus.
OWTTE

e) Explain why it is easier to perform gene therapy on recessively inherited diseases. [2]

a. loss of function mutation → no functional protein is coded for
b. Gene therapy for recessive disease would thus involve introducing a normal allele to express the missing protein either by

Besides gene therapy, there is another field of medical treatment worth noting – organ regeneration using stem cells. Fig. 3.2 below shows a schematic illustration on how organs could be regenerated using tissue engineering and stem cells.

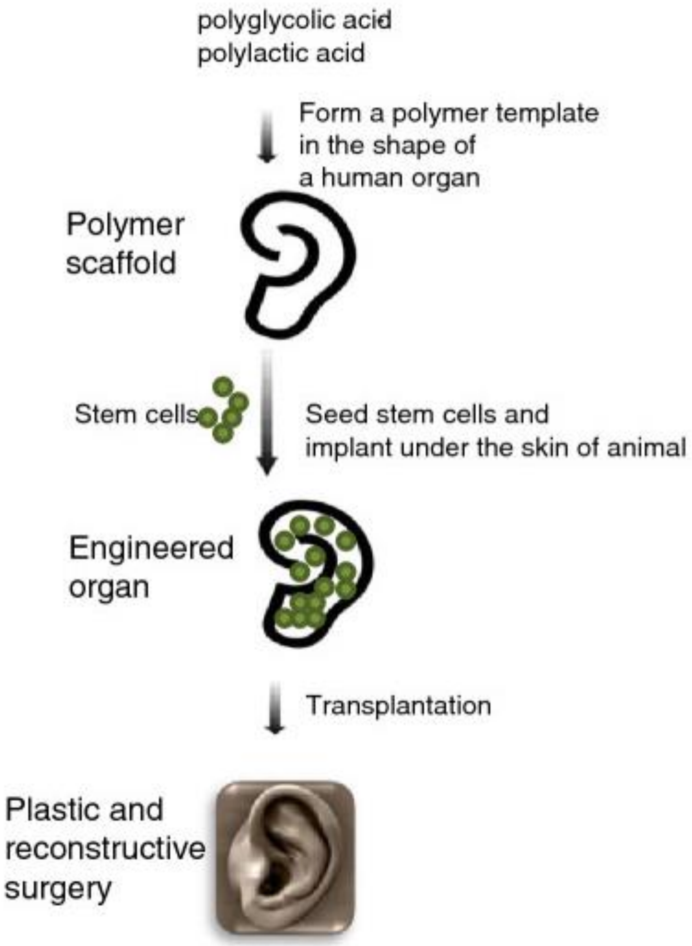


Fig. 3.2

With reference to Fig. 3.2, the stem cells could be isolated from bone marrow and then treated with chemicals to stimulate proliferation so as to form a functional organ.

f) Suggest and explain why the stem cells need to be treated with chemicals to stimulate proliferation. [2]

a. These chemicals are likely to be growth factors,
b. by activating the expression of certain genes

Question 4: Planning Question [12 marks]

Fig. 4.1 shows a colorimeter that can be used to measure the absorbance of particular wavelengths of light by a solution. Different solutes absorb different wavelengths of light. The higher the concentration of a particular solute, the greater corresponding wavelength of light is absorbed.

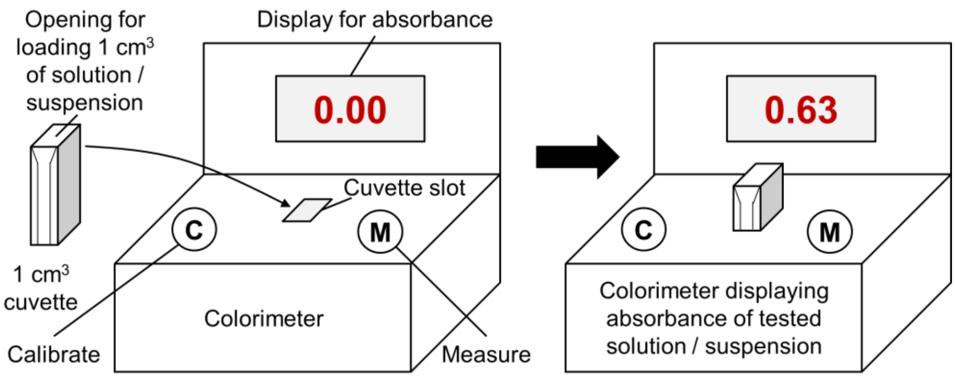


Fig. 4.1

You are required to plan, but not carry out, an investigation into the effect of light intensity on the rate of light dependent reaction of photosynthesis in a leaf extract using the dye DCPIP. When DCPIP is reduced, it changes from blue colour to colourless.

DCPIP (blue) + electrons → reduced DCPIP (colourless)

A leaf extract can be made by mixing finely ground leaf with cold buffer solution. This lead extract should be kept cold and in the dark except when taking samples.

Design an experiment to test the hypothesis that:

The rate of photosynthesis is dependent on the intensity of light.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which *you* must use:

- colorimeter with 1 cm^3 cuvettes
- fresh spinach leaves
- sharp knife
- mortar and pestle

- Cold buffer solution
- 1M $NaHCO_3$
- ice
- light-proof aluminium foil
- DCPIP solution (freshly made)
- 40 W lamp (chosen so it does not generate heat)
- ruler
- fine sharp sand
- a room which can be made dark using curtains
- a variety of different sized beakers, measuring cylinders, syringes and pipettes for measuring volumes

A leaf extract can be made by mixing finely ground leaf with cold buffer solution. This leaf extract should be kept cold and in the dark except when taking samples.

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

<u>Suggested Answer</u>

Aim: To investigate the effect of light intensity on the rate of photosynthesis.

Background:

1. **Effects of light on Photoactivation**
2. **Role of DCPIP in this experiment**
3. **Oxidised DCPIP is dark blue, and when reduced, by high energy electrons from the ETC and hydrogen ions, becomes colourless.**
4. **Hypothesis: The rate of DCPIP reduction and hence the rate of photosynthesis increases linearly with light intensity until light saturation point where rate of photosynthesis will plateau at the maximum rate.**

Procedure & Variables:

1. **A labelled diagram of the experimental set-up**
2. **Procedure should be logical and detailed.**
   a. **Students should be clear about volumes and concentrations used.**
   b. **Variables to be kept constant must be mentioned and**
   c. **control for the experiment as well.**
3. **Table with correct heading columns and units; graph with correct axes decribed or drawn.**

Safety:

Risk	Safety measure	Rationale
1. Sharp knife used in chlorophyll extraction	Care taken in handling of knife	To prevent injury from knife while extracting chlorophyll
2. Irritability of DCPIP	Use of latex gloves and gloves + wash hands thoroughly when handling DCPIP	To prevent allergies and accidental ingestion of DCPIP

Question 5: Free-response Question [20 marks]

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate.
- must be set out in section (a), (b), etc., as indicated in the question.

a) Distinguish between a genomic DNA library and a cDNA library. [5]

	Genomic DNA library	cDNA library
a) Purpose	To study control of gene expression	Isolate a particular gene that is active in a specialized cell and analyse its properties
b) Content	Contains all coding and non-coding sequences	Expressed genes in the tissue – lack non-coding regions
c) Method of construction	Cell's DNA cut using RE and inserted into vectors	mRNA → cDNA using reverse transcriptase which is then inserted into vectors

b) Genetic engineering can also be used for the improvement of crops and animals. Explain, using named examples, the significance of genetic engineering in improving the quality and yield of crop plants and animals. [9]

Improve yield:

1. **Bt corn (named example);**
2. **Bt gene and toxin must be mentioned**
3. **Effects of Bt toxin**
4. **How is crop plant yield improved**

5. **GM salmon / transgenic salmon / AquAdvantage salmon (named example);**

6. **Genes inserted to be mentioned**

7. **Advantage of the genes inserted**
8. **How does gene improve yield of salmon**

Improve quality

9. **Golden rice (named example);**
10. **Details of gene involved**
11. **improve nutritional value of rice**
12. **ref to vit A deficiency**

13. **Enhancing the quality of milk**
14. **Cows have been genetically modified to produce high-protein milk for the cheese industry in New Zealand.**
15. **Name relevant genes involved.**
16. **This should allow cheese-makers to produce more cheese from the same volume of milk.**

A! other valid eg.

c) Discuss the ethical and social issues of (i) GMOs, (ii) stem cells and (iii) gene therapy. [6]

GMOs (G)

Ethical

1. **Tampering with nature / "playing God" by mixing genes among species; genetic modifications of animals/plants especially resulting in enhancement; draws parallels that human genetic modification for enhancement one day may occur; -> integrity of species is not presserved**

Social

2. **Environmental protection from GM plants; safeguards of products with GM components may not be adequate; to prevent transfer of genes/pollen from GM organisms to wild-type organisms;**

STEM CELLS (S)

Ethical

1. **The questions at the centre of the controversy concern the nature of early human life and the legal and moral status of the human embryo**
2. **Some are concerned about the potential misapplication of nuclear transfer for reproductive cloning purposes.**

Social

3. **Cost of treatment is high therefore it is only accessible to the rich.**

GENE THERAPY (T)

Ethical

4. Tampering with human genes in any way will inevitably lead to the practice of eugenics;

Social

5. Gene therapy is costly; Only the rich have access to it and not the poor;